Entropy Meets Importance: A Unified Head Importance–Entropy Score for Stable and Efficient Transformer Pruning

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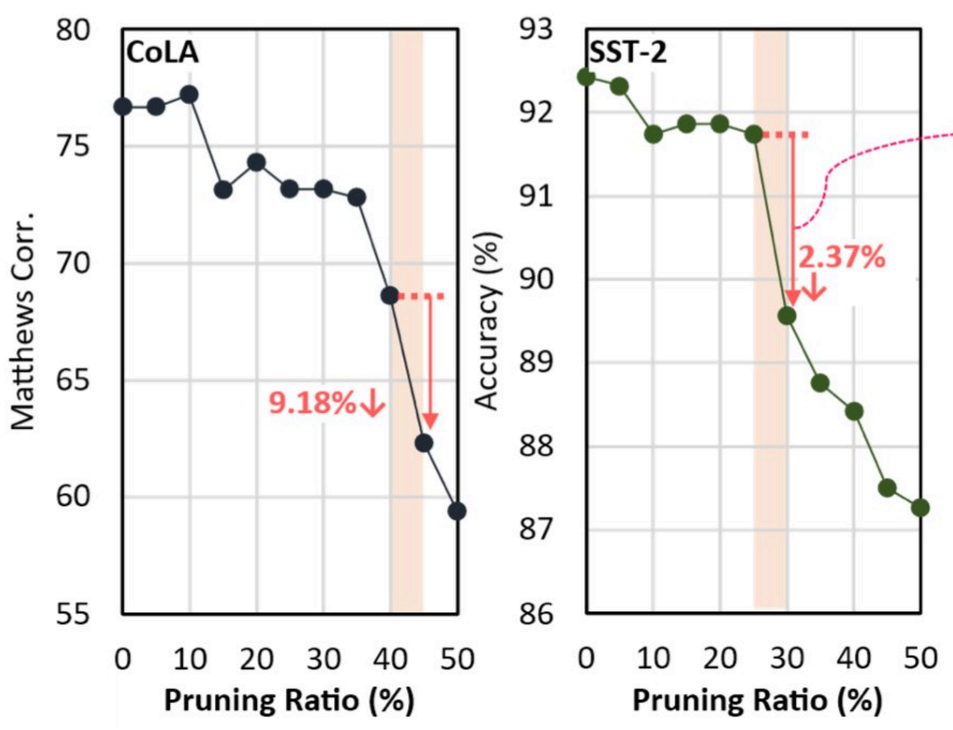
Overview

Transformer-based models continue to **scale in layers and attention heads, improving accuracy and long-range reasoning** **but significantly increasing inference cost**

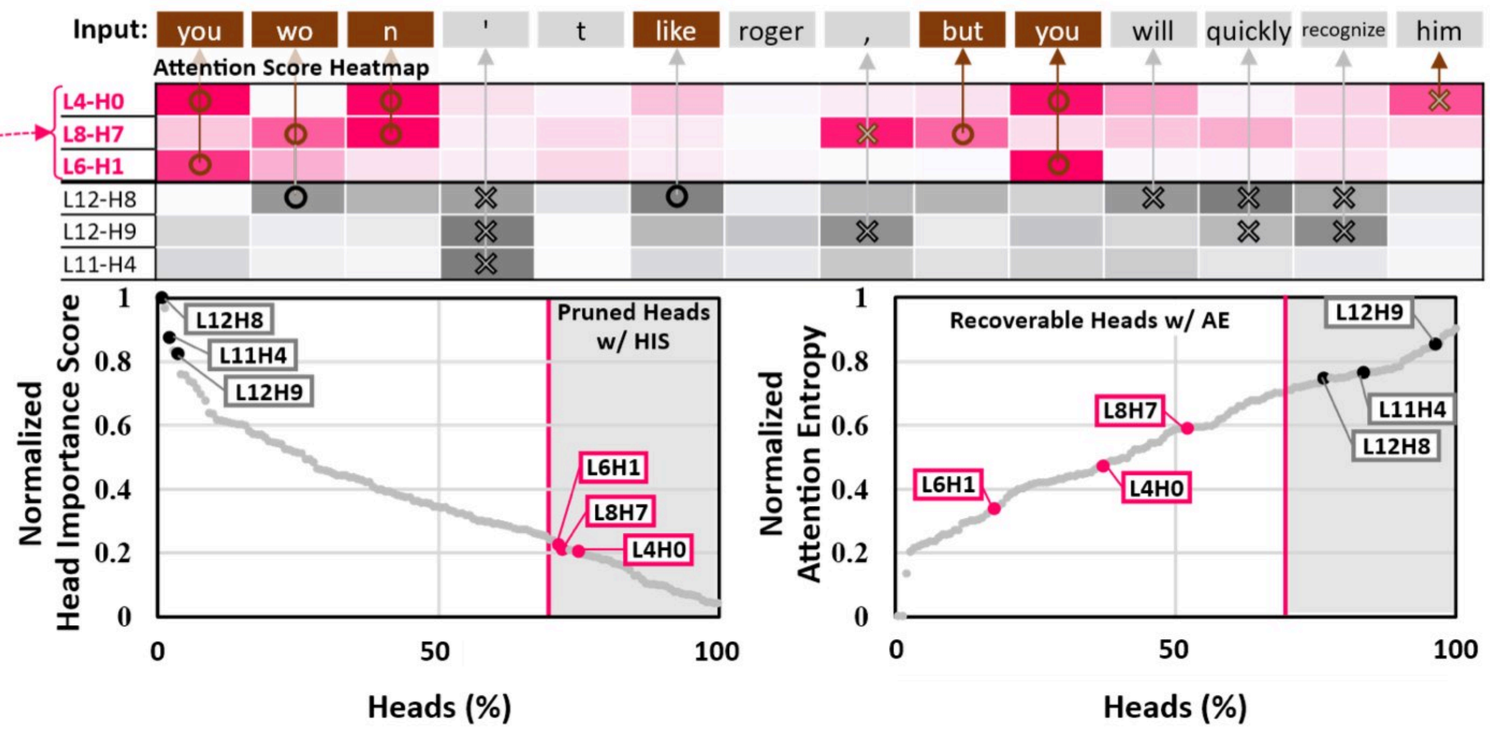
- **Gradient-based Head Importance Scores (HIS):** widely used for pruning but are limited, capturing only gradient contribution and ignoring attention pattern diversity
- **Head Importance-Entropy Score (HIES):** combines HIS with attention entropy

➔ **Accuracy:** up to **17.62%**, **Stability:** up to **2.04×** over HIS-only methods.

Motivation



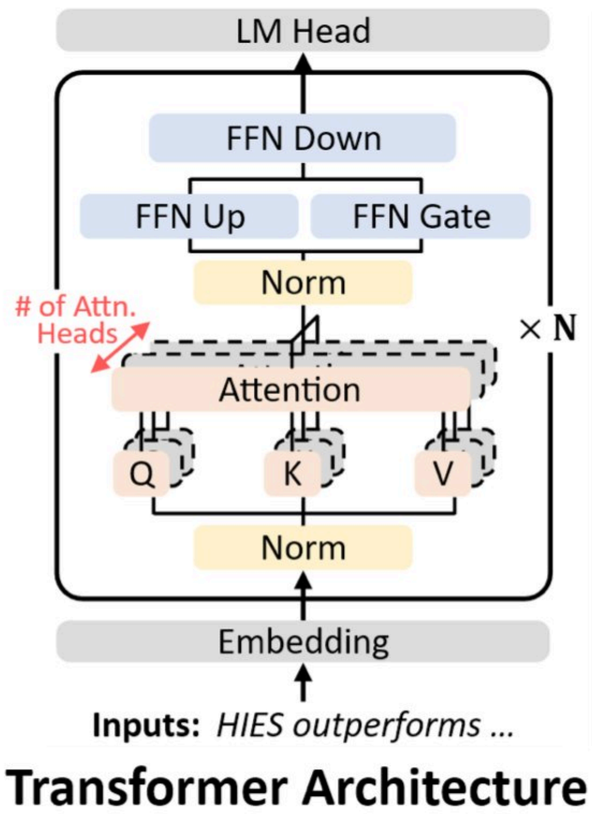
Accuracy *Sharp-Drop* Regime



Analysis of attention heads in accuracy sharp-drop regions

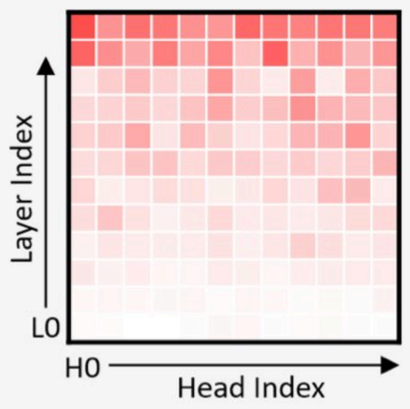
Attention Entropy (AE) can serves as an indicator of stability, mitigating accuracy sharp-drops!

Head Importance-Entropy Score (HIES)



Transformer Architecture

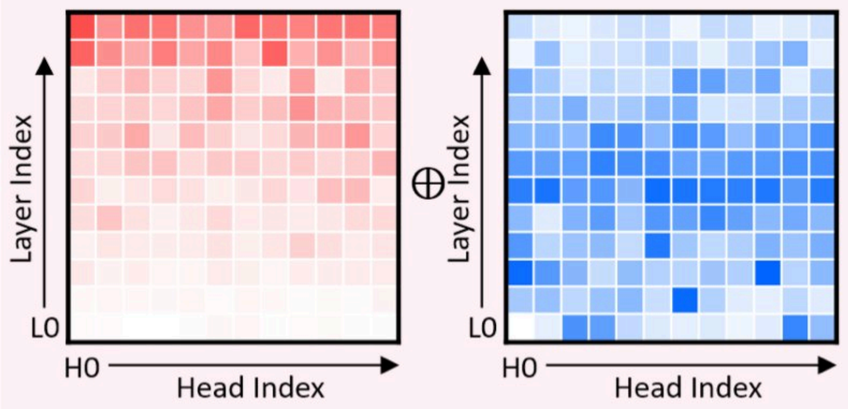
HIS: First-order loss-based



$$\mathrm{HIS}_{l,h} = \mathbb{E}_{x\sim\mathcal{D}}\left|\partial \mathrm{A}_{l,h}(x)^{\mathrm{T}} \frac{\partial \mathcal{L}(x)}{\partial \mathrm{A}_{l,h}(x)}\right|$$

HIS

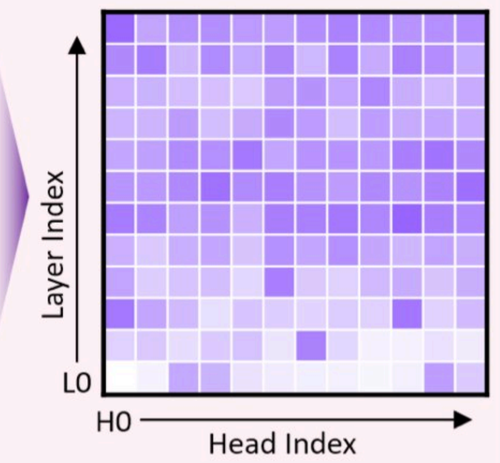
HIS: First-order loss-based **AE:** Stability



$$\mathrm{AE}_{l,h} = -\sum_{i=1}^{n} p_i^{(l,h)} \log p_i^{(l,h)} \ \text{s.t.} \sum_{i=1}^{n} p_i^{(l,h)} = 1$$

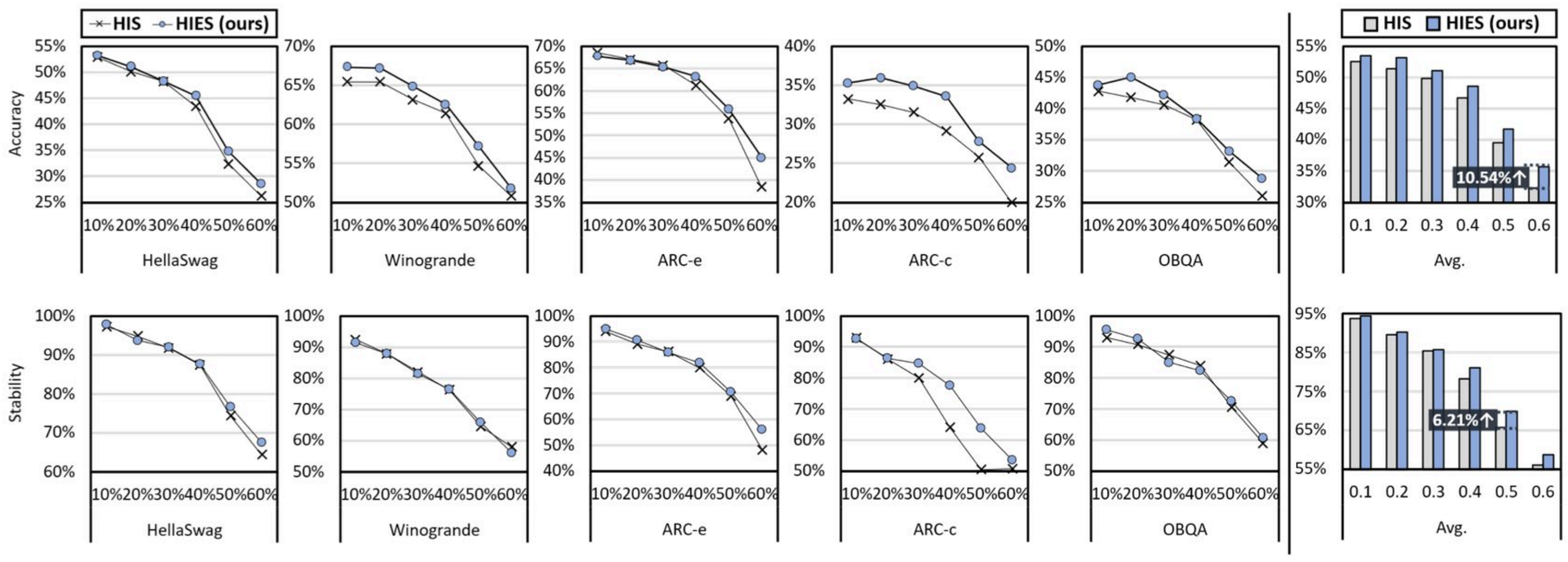
HIES (Ours)

HIES for *Higher Task Accuracy* and *Stability*



$$\mathbf{HIES}_{l,h} = \alpha\widehat{\boldsymbol{HIS}_{l,h}} + (1-\alpha)(1-\widehat{\mathbf{AE}_{l,h}})$$

Evaluation



HIES improves accuracy by +10.54% and stability by +6.21% compared to HIS

For details on the **Theoretical Analysis** (risk decomposition, orthogonality proof, stability analysis) and **Further Experiments** (on NLP, VQA, and other tasks), please refer to the paper.

Paper **Lab**